Curriculum Vitae
Materials Science and Engineering

Yeeun Lee

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RESEARCH INTERESTS

Machine-Learning Interatomic Potentials, Multiscale Molecular Simulation, Molecular Coarse-Graining, Energy Materials, Solid State Electrolytes, Electrochemical Interfaces.

RESEARCH EXPERIENCE

Graduate Research **Apr. 2024 - Present**

1) Technische Universität München (TUM), Germany ***Advisor: Prof. Patrick Rinke***

- **Project: Machine-Learning Molecular Dynamics of Electric-Field-Induced Vibrational Dynamics**
  - Performed molecular dynamics simulations under external electric fields using pretrained MACE machine-learning interatomic potentials in Linux/Slurm HPC environments.
  - Analyzed MD trajectories using Python to extract time-dependent dipole responses, IR spectra, and mode-resolved vibrational energies under different electric-field conditions.
  - Investigated field-induced vibrational behavior and energy transfer and benchmarked ML-MD results against DFT reference calculations and experimental observations.

2) Forschungszentrum Jülich, Germany ***Advisor: Dr. Baohu Wu***

- **Project: Operando Investigation of Structural and Interfacial Evolution in PVDF-HFP-Based Polymer Electrolytes for Li Solid-State Batteries**
  - Synthesized PVDF-HFP/Succinonitrile-based polymer electrolytes and prepared operando-cell configurations
  - Combined in situ Raman spectroscopy, SAXS/WAXS, operando battery cycling, and electrochemical testing to correlate structural evolution with electrochemical behavior.

3) Technische Universität München (TUM), Germany ***Advisor: Prof. Jennifer L. M. Rupp***

- **Project 1: Electrochemical and Thermal Analysis of Prussian Blue Analogue (PBA) Thin Films for Thermobatteries**
  - Investigated the electrochemical and thermal behavior of PBA thin films for thermobattery applications.
- **Project 2: WO_3 Thin Films via Spray Pyrolysis for Proton-Based Energy Storage**
  - Synthesized WO_3 thin films by spray pyrolysis while systematically varying substrate temperature and spray volume to control film morphology and thickness.
  - Correlated SEM, Raman spectroscopy, XRD, cyclic voltammetry, and EIS results to establish processing–structure–electrochemical property relationships.
  - Identified strong processing-dependent variations in film thickness, proton intercalation, and charge-transfer resistance, and determined deposition conditions that promoted reversible proton intercalation/deintercalation and stable electrochemical performance.

Undergraduate Research **Mar. 2023 - Sep 2023**

1) Korea University, Seoul, Korea ***Advisor: Prof. Seok Su Sohn***

- **Project: Hydrogen embrittlement behavior in CoCrFeMnNi high-entropy alloys (HEAs) under cryogenic condition**

Research Intern **May. 2022 - Jan. 2023**

1) SK On, Joint Battery Platform Project with **Korea University**

- **Project: Smart Battery & EV Charging Platform Development**

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EDUCATION

M.S. in Materials Science & Engineering **Sep. 2026**
Technical University of Munich (TUM), (Munich, Germany)

Overall GPA: 1.6 (German grading scale; 1.0 = best; 5.0 = fail)

B.S. in Materials Science & Engineering **Feb. 2024**
Korea University, (Seoul, Korea)

PUBLICATION

†(co-)first author, *(co-)corresponding author

1. **Y. Lee**, B. Wu*. Operando Investigation of Structural and Interfacial Evolution in PVDF-HFP/Succinonitrile Polymer Electrolytes. *(In preparation)*
2. **Y. Lee**†, N. Isaev†, J. L. M. Rupp*. Proton Intercalation in WO_3 Thin Films Synthesized by Spray Pyrolysis under Varying Parameters. *(In preparation)*
3. P. Borah†, **Y. Lee**, A. Bharati*. Small-Angle Neutron Scattering Analysis of Pectin Conformation in Solution. *(In preparation)*
4. N. Bhatia†, **Y. Lee**†, P. Rinke*. Machine-Learning Molecular Dynamics of Electric-Field-Induced Vibrational Dynamics. (*In preparation)*

PRESENTATIONS

1. N. Isaev, **Y. Lee**, K. Bohnet, J. L. M. Rupp*. "Tunable Proton Intercalation in Tungsten Oxide Thin Films for Proton-Based Energy Storage Devices." E-MRS 2026 Spring Meeting, Strasbourg, France. May 25-29, 2026.
2. **Y. Lee**, A. Bharati, B. Wu*. "Operando Structural Evolution of PVDF-HFP Polymer Electrolytes in Lithium Batteries." 7th International Energy Storage Materials Conference, Dresden, Germany. Sep 20-23, 2026.
3. **Y. Lee**, B. Wu*. "Co-Solvent-Engineered PVDF-HFP/Succinonitrile Quasi-Solid Electrolytes: Linking Residual Solvent and Nanostructure to Li Deposition via Operando SAXS/WAXS." 2026 Fall Meeting of the Korean Electrochemical Society (KECS), Jeju, Korea. Nov 18-20, 2026.
4. **Y. Lee**, B. Wu*. "Operando Structural Evolution of PVDF-HFP Polymer Electrolytes in Lithium Batteries." MLZ User Meeting 2026, Munich, Germany. Dec 2-3, 2026.

SCHOLARSHIPS

- **Merit-based Waiver Scholarship for Academic Excellence** **Oct. 2025 - Sep. 2026**
  Technical University of Munich (TUM)
  - Awarded to the top-performing student in the department based on academic excellence

HONORS & AWARDS

- **Capstone design Award**, *Korea university* **Sep. 2023**
- **Excellence Award**, *SK-On Battery Platform Research* **Feb. 2023**
- **The Grand Award,** *AI model for predicting solar panel power generation contest.* **July. 2023**
- **Outstanding Organic Chemistry Mentoring Award**, *korea university* **Mar. 2022**

EXTRACURRICULAR ACTIVITIES

- **AI Solar Power Prediction Contest**, *Korea University* **Feb. 2023 - July. 2023**
  - Developed a stacking ensemble for solar power forecasting, integrating tree-based and deep-learning models including LightGBM, Random Forest, MLP, and LSTM-based architectures.
  - Evaluated and optimized predictive performance for time-series energy forecasting, achieving 1st Place in the competition.

VOLUNTEER ACTIVITIES

- **Community Support Program for the Visually Impaired** **Mar. 2022 - Aug. 2023**
  - Assisted visually impaired individuals through community service activities, promoting accessibility and inclusion.

- **Educational Mentoring Program for Underprivileged Students** **Nov. 2021 - Feb. 2022**
  - Provided academic mentoring and educational support for students in marginalized communities, aiming to promote equal learning opportunities and address educational disparities.

SKILLS

- **Computational/Programming**
  - Python, Linux, Slurm, Matlab, Machine Learning, Molecular dynamics
- **Electrochemical & Battery Testing**
  - Battery cell assembly, Operando battery cycling, BioLogic EC-Lab
- **Materials Characterization**
  - In-Situ Raman, SAXS/WAXS, XRD, AFM, SEM, DSC, TGA, FTIR, EBSD, Tensile testing
- **Materials Processing & Synthesis**
  - Tape casting, Solution casting, Spray pyrolysis, Electrochemical deposition, Glovebox handling